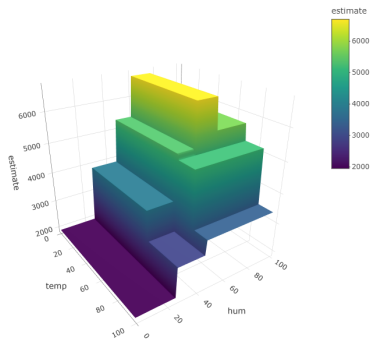
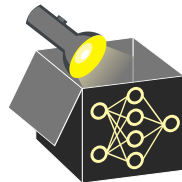


Interpretable Machine Learning

Interpretable Models 1

Rule-based Models



Learning goals

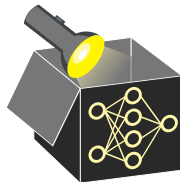
- Decision trees
- RuleFit
- Decision rules

DECISION TREES

► "Breiman et al." 1984

Idea: Partition data into axis-aligned regions via greedy search for feature cut points (minimizing a split criterion), then predict a constant mean c_m in each leaf region $\mathcal{R}_m$:

$$\hat{f}(x) = \sum_{m=1}^M c_m \mathbb{1}_{\{x \in \mathcal{R}_m\}}$$



DECISION TREES

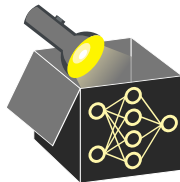
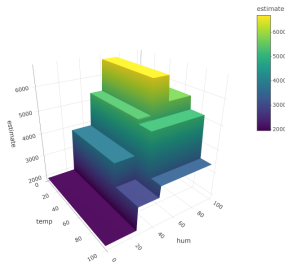
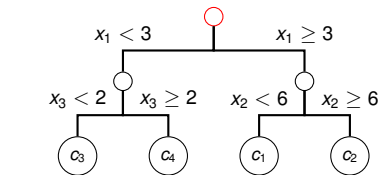
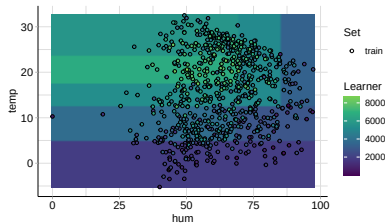
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$\mathcal{R}_m$:

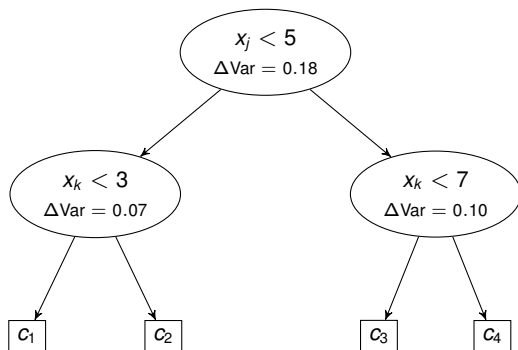
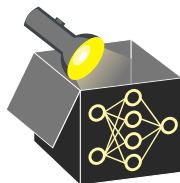
$$\hat{f}(x) = \sum_{m=1}^M c_m \mathbb{1}_{\{x \in \mathcal{R}_m\}}$$

- Applicable to regression and classification
- Models interactions and non-linear effects
- Handles mixed feat, spaces & missing values



INTERPRETATION OF TREE-BASED MODELS

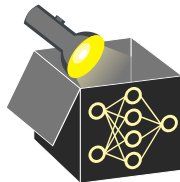
- Interpretation via path of decision rules along tree branches
- **Feature importance** (quantifies how often and how usefully x_j is used):
 - For each split on feature x_j , record the decrease in the split criterion
 - Aggregate this over the tree: sum or avg. over all splits involving x_j
 - Split criterion: variance (regression), Gini index / entropy (classif.)



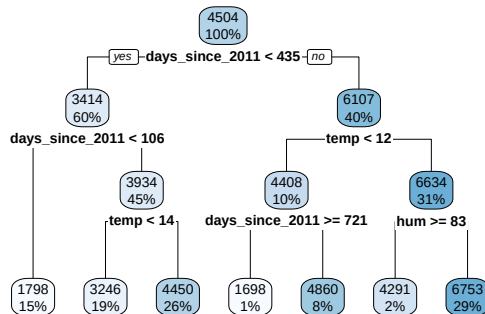
- Each ΔVar is assigned to the splitting feature
- Feature importance = sum of all ΔVar for that feat.:
 - x_j : 0.18
 - x_k : $0.07 + 0.10 = 0.17$

DECISION TREES - EXAMPLE

- Fit decision tree with tree depth of 3 on bike data
- E.g., mean prediction for the first 105 days since 2011 is 1798
 ↪ Applies to 15% of the data (leftmost branch)
- `days_since_2011`: highest feat. importance (explains most of variance)



Feature	Importance
<code>days_since_2011</code>	79.53
<code>temp</code>	17.55
<code>hum</code>	2.92



UNBIASED RECURSIVE PARTITIONING

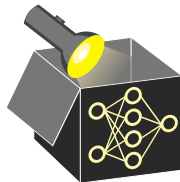
► "Hothorn" 2006

► "Zeileis" 2008

► "Strobl" 2007

Problems with CART (Classification and Regression Trees):

- 1 Selection bias towards high-cardinal/continuous features
- 2 Splits on any improvement, regardless of significance
~> prone to overfitting



UNBIASED RECURSIVE PARTITIONING

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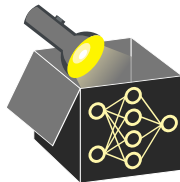
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Unbiased recursive partitioning via conditional inference trees (`ctree`) or model-based recursive partitioning (`mob`):

- ❶ Separate selection of **feature used for splitting** and **split point**
- ❷ Hypothesis test as stopping criteria



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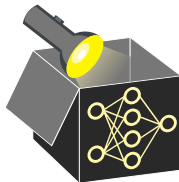
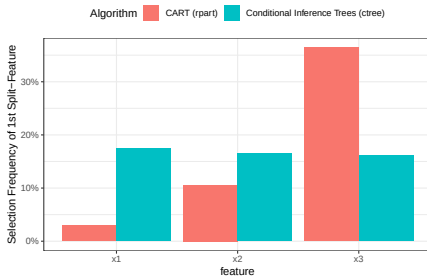
- 1 Separate selection of **feature used for splitting** and **split point**
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Example (selection bias):

Simulate data ($n = 200$), $Y \sim N(0, 1)$ and 3 features of different cardinality indep. from Y (repeat 500 times):

- $X_1 \sim \text{Binom}(n, \frac{1}{2})$
- $X_2 \sim M(n, (\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}))$
- $X_3 \sim M(n, (\frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}))$

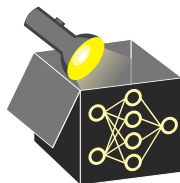
Which feature is selected in the first split?



UNBIASED RECURSIVE PARTITIONING

Differences to CART:

- Two-step approach (finds 1. most significant split feat., 2. best split point)
- Parametric model (e.g. LM instead of constant) can be fitted in leaf nodes
- Significance of split (p-value) given in each node
- `ctree` and `mob` differ in hypothesis test used for selecting the split feature (independence test vs. fluctuation test) and how to find the best split point

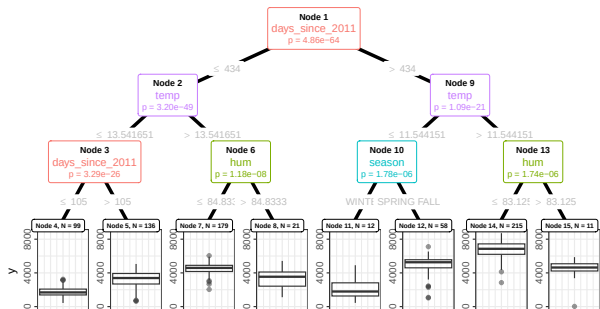


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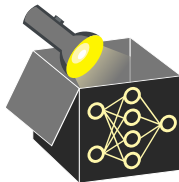
Example (`ctree`): Bike data (constant model in final nodes)



Train MSE:

758,844 (`ctree`)

742,244 (`mob`)

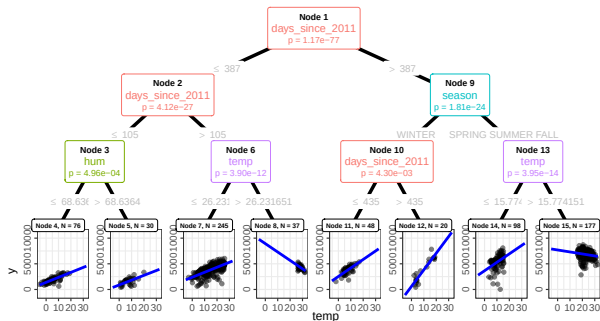


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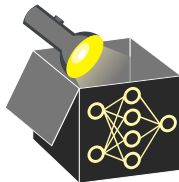
Example (`mob`): Bike data (linear model with `temp` in final nodes)



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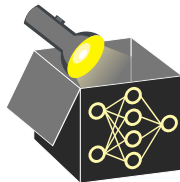
OTHER RULE-BASED MODELS

Decision Rules

► "Holte" 1993

- Flat list of simple "if – then" statements
 $\rightsquigarrow$ very intuitive and easy-to-interpret
- Mainly devised for classification
 (support for regression is limited)
- Numeric features are typically discretised

```
IF  $x_1 \leq 2.3$  AND  $x_4 = \text{"A"}$  THEN  $y = 1$   
ELSE IF  $x_2 > 5.0$  THEN  $y = 2$   
ELSE  $y = 3$ 
```

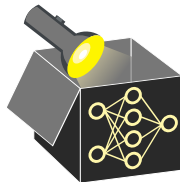


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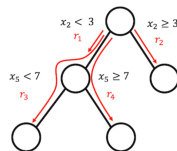
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RuleFit ► "Friedman and Popescu" 2008

- Extract binary rules $r_m(\mathbf{x}) \in \{0, 1\}$ from many shallow trees (one per root-to-leaf path)
- Fit an L_1 -regularized LM
$$\hat{f}(\mathbf{x}) = \beta_0 + \sum_m \beta_m r_m(\mathbf{x}) + \sum_j \gamma_j x_j$$
- Regularization retains only a few rules
⇒ sparse, non-linear, interaction-aware
- Coefficients relate to rule/feature importance



► "Molnar" 2022